SAMANTHEX

Sanger Consensus Assembler

Version 3.0 - User Manual

Gabriel Dias Moreira

Universidade Federal de Minas Gerais (UFMG)

Graduation Program in Microbiology

1. Introduction

SAMANTHEX is a bioinformatics tool designed to assemble high-quality consensus sequences from Sanger sequencing data. It provides an intuitive graphical interface for processing chromatogram files, with integrated quality trimming, multi-replicate consensus assembly, and NCBI BLAST search.

The tool accepts ABI (.ab1) and PHD (.phd.1) file formats, supports multiple forward and reverse replicates per sample, and uses Phred quality scores for intelligent base calling during consensus generation.

Key Features

- Load multiple forward and reverse read files (replicates)
- Automatic quality trimming using Mott's algorithm (same as phred)
- Quality-aware consensus with Phred score boosting for concordant bases
- Configurable minimum Phred quality threshold (10-40)
- NCBI BLAST integration with selectable database and program
- Per-base quality visualization with Q20/Q30 reference lines
- Detailed quality statistics (length, GC%, mean Q, %Q20, %Q30, N count)
- Export: FASTA, quality report, BLAST results
- Standalone executable - no Python installation required

2. Installation

Option A: Standalone Executable (Recommended)

Download the Samanthex.exe file from the releases page. No installation is required - simply double-click to run. This is the recommended option for users who do not have Python installed.

Option B: From Source Code

Requirements: Python 3.9 or later with tkinter.

```
# Clone the repository
git clone https://github.com/Dias0202/SAMANTHEX_2.0.git
cd SAMANTHEX_2.0

# Install dependencies
pip install -r requirements.txt

# Run the application
python run.py
```

Building the Executable Yourself

```
pip install pyinstaller
python build_exe.py
```

The executable will be created in the dist/ directory.

3. Usage Guide

Step 1: Select Input Files

Click "Select Forward Files" to load your forward sequencing reads, and "Select Reverse Files" for reverse reads. You can select multiple files per direction - these are treated as replicates of the same sample.

Supported formats: ABI chromatogram files (.ab1) and Phred output files (.phd.1). You can mix formats within the same direction.

Note: You can use only forward OR only reverse files if bidirectional sequencing is not available.

Step 2: Configure Settings

Setting	Description	Default
Sequence Name	Identifier for the output	Auto timestamp
Min. Phred	Quality threshold (bases below = N)	20
Trim ends	Enable Mott's quality trimming	Enabled

Step 3: Assemble Consensus

Click "Assemble Consensus" to start the pipeline. The progress bar shows the current stage. Results are displayed in three tabs:

- Consensus Sequence - The final FASTA output with a 'Copy' button.
- Quality Report - Detailed statistics including length, GC content, quality distribution, trimming details, and warnings.
- BLAST Results - Populated after running a BLAST search.

Step 4: Run BLAST (Optional)

After assembly, click "Run BLAST" to search the consensus against NCBI databases. You can configure the database (nt, refseq_rna, etc.), program (blastn, megablast), and number of hits. An internet connection is required. BLAST queries typically take 1-5 minutes.

Step 5: Export Results

Click "Export Results" to save all outputs to a chosen directory: the consensus FASTA, quality report, and BLAST results (if available).

4. Assembly Pipeline

Overview

```
1. READ    - Load AB1/PHD files
2. TRIM    - Remove low-quality ends (Mott's algorithm)
3. MERGE F - Merge forward replicates (quality-weighted)
4. MERGE R - Reverse-complement + merge reverse replicates
5. MERGE   - Align and merge forward + reverse consensuses
6. FILTER  - Apply Phred threshold (low quality -> N)
7. REPORT  - Compute statistics and display results
```

Quality Trimming: Mott's Algorithm

Sanger sequencing reads typically have low-quality bases at both ends. Mott's algorithm finds the contiguous region that maximizes the cumulative sum of (Qi - threshold) across all positions. This is mathematically equivalent to Kadane's maximum subarray algorithm.

This is the same approach used by phred, one of the most widely cited programs in bioinformatics.

Reference: Ewing B, Green P. Base-calling of automated sequencer traces using phred. II. Error probabilities. Genome Research. 1998;8(3):186-194.

Quality-Aware Consensus

During sequence merging, quality scores determine base selection:

- Matching bases: Quality is boosted (Q = Q1 + Q2, capped at 93). Agreement from independent reads increases confidence.
- Mismatching bases: The base with the higher Phred score is selected.
- Gaps: The non-gap base is used with its original quality.
- N-substitution is deferred to the final step to prevent premature information loss during intermediate merges.

Alignment Parameters

Parameter	Value	Rationale
Mode	Local	Finds best overlap region
Match	+2	Standard reward for correct calls
Mismatch	-1	Mild penalty (errors expected)
Gap open	-5	Prevents spurious gaps
Gap extend	-2	Accommodates real indels

5. Interpreting Results

Quality Statistics

Metric	Good Value	Description
Length	>400 bp	Total consensus length
Mean Quality	>30	Average Phred score
%Q20	>95%	Bases with Q>=20 (99% accuracy)
%Q30	>80%	Bases with Q>=30 (99.9% accuracy)
N bases	<2%	Bases below quality threshold
GC Content	40-60%*	Depends on organism

* Expected GC content varies by organism and gene.

Quality Profile Visualization

The Quality Report tab displays a per-base quality bar chart with color coding:

- Green bars: Quality >= 30 (excellent, 99.9% base call accuracy)
- Yellow bars: Quality 20-29 (good, 99% accuracy)
- Red bars: Quality < 20 (poor, consider reviewing)

Dashed lines indicate Q20 and Q30 thresholds. A high-quality consensus should have predominantly green bars.

BLAST Results

For each BLAST hit, the following information is shown:

- Title: Full description of the matching database sequence
- Score: Alignment bit score (higher = better match)
- E-value: Statistical significance (lower = more significant)
- Identity: Percentage of matching bases in the alignment
- Alignment: Visual display of query vs. subject sequences

6. Troubleshooting

Common Issues

High N content in consensus

If the consensus has many N bases, try: (1) lowering the Phred threshold, (2) checking input file quality in the original sequencing software, (3) ensuring forward and reverse reads are from the same sample.

Short consensus after trimming

If reads are very short after trimming, the raw data may be of poor quality. Try disabling trimming to see the full (untrimmed) consensus and evaluate the quality profile.

BLAST timeout or connection error

BLAST queries require an internet connection. If you experience timeouts, try again later - NCBI servers may be under heavy load. Consider using megablast for faster searches.

Application does not start (executable)

Some antivirus software may flag PyInstaller executables. Add an exception for Samanthex.exe in your antivirus settings. The application is safe and open-source.